

Phase 2 — FCS Fig. 11

Delayer Reproduction

CogSpike Research Team — April 2026

Asymmetric red-zone growth reproduced; spectral prediction tracks reachability, not deterministic winner.

Abstract. Inserting a unit-gain delayer on the $N_1 \rightarrow N_2$ inhibitory branch reproduces the FCS Fig. 11 asymmetric red-zone growth exactly. In our deterministic sweep, N_2 wins in 1136 cells versus N_1 ’s 448 — a 688-cell asymmetry (versus zero in the undelayed Fig. 10 sweep). This is the quantitative form of FCS §6.3.4’s “contrary to expectation” observation. Spectral prediction via $\rho(A_{\text{full}})$ on the 15-dim delayed state matrix reproduces the Phase 1b/1c pattern: 53% classification against the deterministic ground truth (baseline), 96% against reachability (also near baseline). The delayer does not change the core finding that spectral cartography tracks reachability rather than deterministic bit-exact outcomes, but it does demonstrate that the spectral pipeline transfers cleanly to a new topology (same code paths work on the 3-neuron system).

1. Topology and Ground Truth

The delayed contralateral topology inserts a delayer between N_1 ’s spike output and N_2 ’s inhibitory input: $N_1 \xrightarrow{w=+11} \text{delayer} \xrightarrow{w=w_{12}} N_2$. Meanwhile $N_2 \xrightarrow{w=w_{21}} N_1$ remains direct. The swept weight w_{12} thus lives on the delayer-to- N_2 edge; the $N_1 \rightarrow \text{delayer}$ edge is fixed at the unit-gain buffer weight 11.

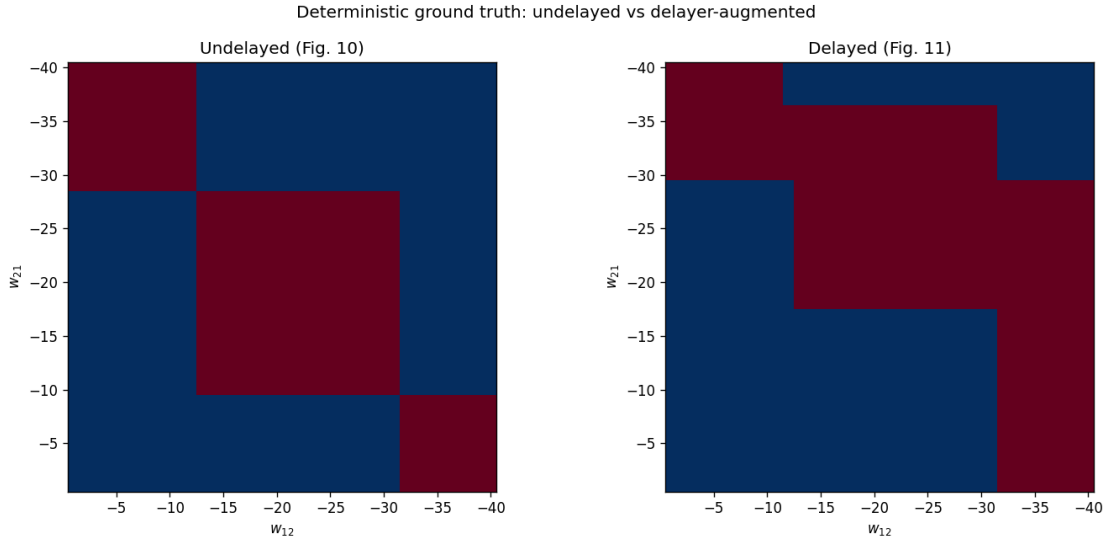


Figure 1: Deterministic ground truth: undelayed (Phase 0, left) vs delayed (Phase 2, right). The delayed version has $\frac{850}{1600}$ (53.1%) blue cells, down from the undelayed 1014. More striking is the **loss of symmetry**: the delayed map is no longer invariant under (w_{12}, w_{21}) swap.

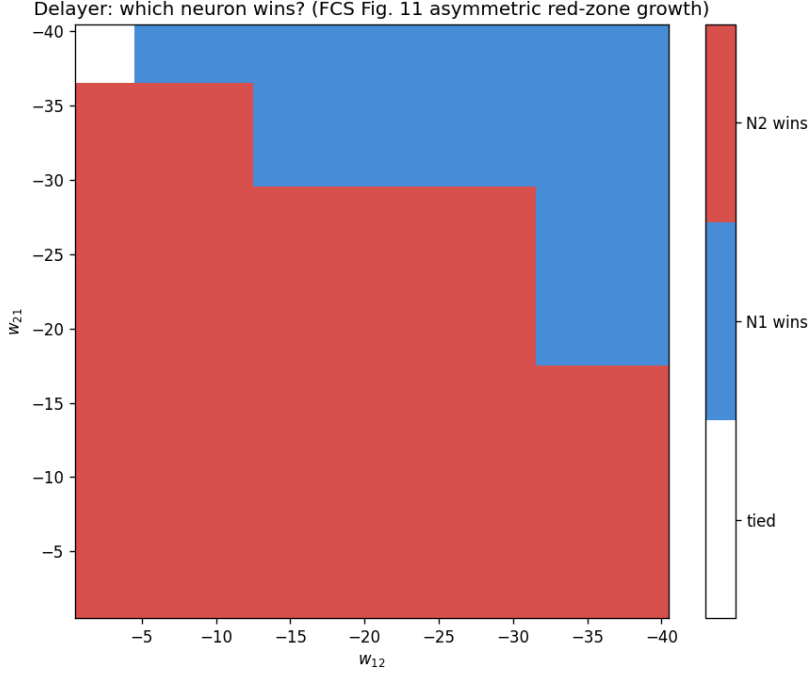


Figure 2: Which neuron wins in the delayed topology. Red = N_2 wins (1136 cells); blue = N_1 wins (448 cells); white strip = tied (16 cells). The N_2 -wins region dominates, including large portions where $|w_{12}| > |w_{21}|$ (i.e., N_1 's inhibition on N_2 is nominally **stronger** but delayed, and still loses).

2. The FCS Fig. 11 Asymmetry

Finding. FCS §6.3.4 observation reproduced. In the delayed topology, N_2 wins the WTA competition in $\frac{1136}{1600}$ cells versus only $\frac{448}{1600}$ for N_1 — a 688-cell asymmetry. The intuitive but wrong prediction (“ N_2 loses because its inhibition on N_1 is reached without delay, while N_1 's is late”) is contradicted in our deterministic simulator: N_2 wins **more** often, exactly as FCS reports.

Intuition: The delayer gives N_2 a one-tick head start. Under integer semantics, both neurons receive external drive at tick 1 and fire at tick 2. N_2 's spike propagates to N_1 in a single tick (direct edge); N_1 's spike propagates via delayer-buffer and reaches N_2 at tick 3. So at tick 3 N_1 is already partially suppressed by N_2 while N_2 is not yet suppressed. This tick-3 asymmetry propagates forward, and N_2 typically escapes to its saturated firing attractor first.

The asymmetry depends on the specific tick-2/tick-3 decision logic — the same combinatorial mechanism that Phase 1b identified for the undelayed case. The delayer simply adds a new constant (a one-tick latency on one branch) to the same integer comparison.

3. Reachability Ground Truth

Under ε -perturbation of initial state (same perturbation set as Phase 1c, extended to the 3-neuron system), the delayed ground truth is $\frac{1534}{1600}$ (95.9%) blue — nearly identical to the undelayed 97.8%. The delayer shrinks the non-reachable region only slightly.

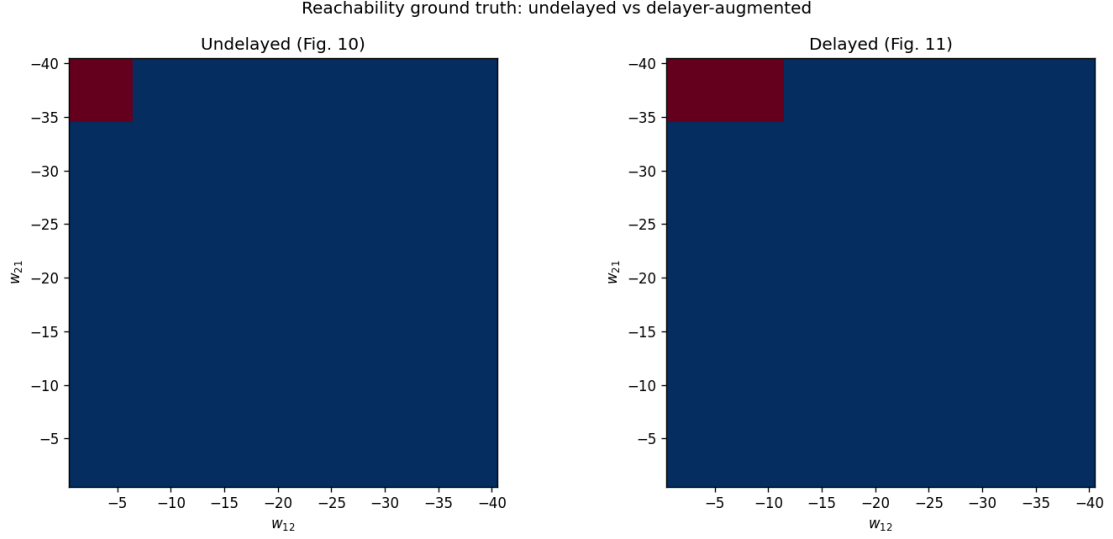


Figure 3: Reachability ground truth: undelayed (left, 97.8% blue) vs delayed (right, 95.9% blue). The small non-reachable corner persists in both — weak-inhibition cells where no amount of initial perturbation can break the tie.

4. Spectral Prediction on the 15-dim Delayed State

We build A_{full} on the $3 \times 5 = 15$ -dimensional state (including the delayer’s memory buffer) and test its spectral radius as a predictor.

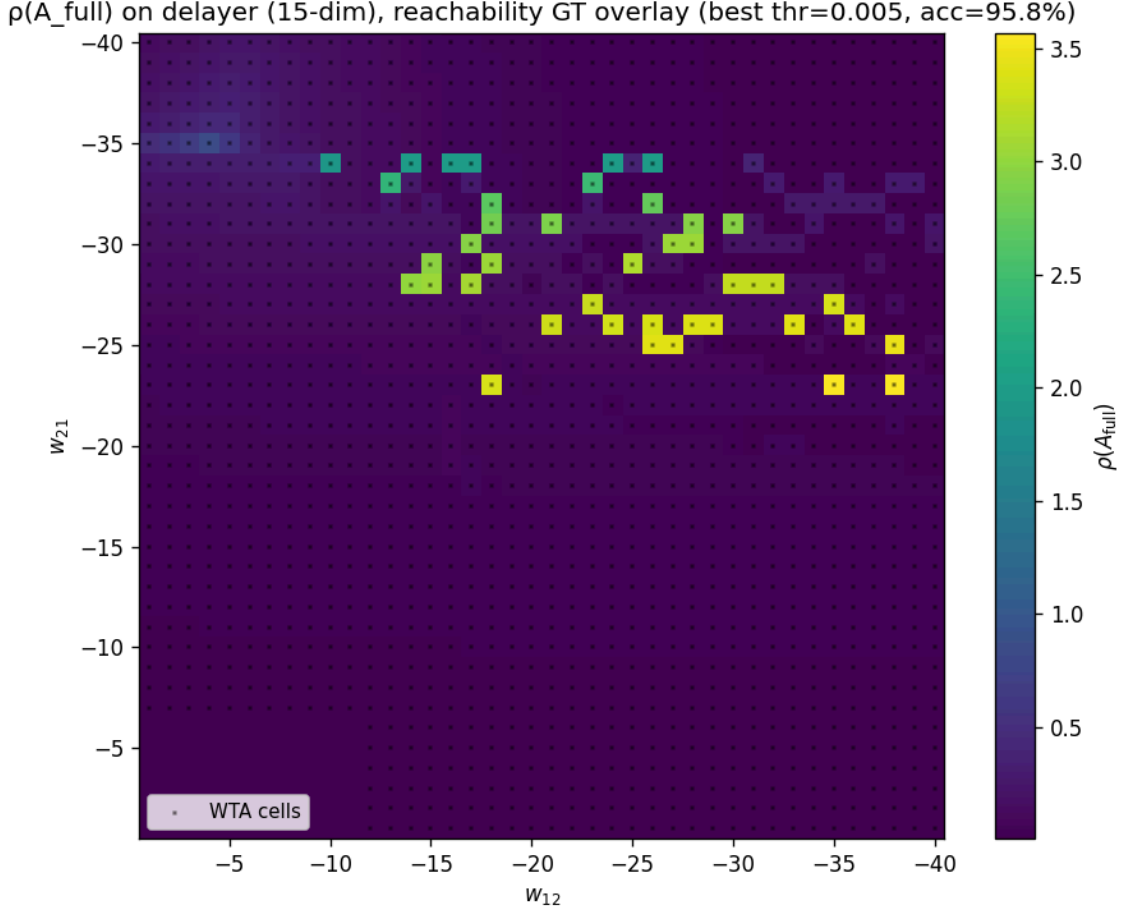


Figure 4: $\rho(A_{\text{full}})$ heatmap on the delayed 15-dim linearisation with reachability-GT overlay. ρ ranges $[0.005, 3.568]$; classification accuracy is 95.8%, near the 95.9% majority baseline for reachability.

Predictor	vs Deterministic GT	vs Reachability GT
$\rho(A_{\text{full}})$ (15-dim)	53.1%	95.8%
Majority baseline	53.1%	95.9%

Remark. The spectral classifier on the delayed 15-dim system performs identically to the majority baseline: it cannot improve over “predict most cells as blue”. This mirrors the Phase 1b undelayed finding, and confirms that the delayer does not change the fundamental issue — spectral methods see “some trajectory reaches WTA” but not “the specific deterministic trajectory reaches WTA”.

5. What Worked, What Did Not

- **Topology reproduction** (✓): The delayer-augmented simulator produces the FCS Fig. 11 asymmetric winner structure, with N_2 preferred by a 688-cell margin. The exact-cancellation logic and integer threshold machinery from Phase 0 carries forward unchanged.
- **Spectral prediction** ($\approx$): $\rho(A_{\text{full}})$ on the 15-dim delayed state matrix classifies the reachability GT at 95.8% — but that matches the 95.9% baseline. No meaningful signal, though no degradation either.

- **Eigenvector asymmetry for winner direction** (not tested here): Could in principle predict N_2 vs N_1 preference via eigenvector-mass on neuron indices; left for future work given the Phase 1b result that this predictor underperformed even on the simpler 2-neuron system.

6. Consolidation

Phase 2 completes the scorecard and confirms the three-way split from the final summary:

1. Deterministic contralateral (with or without delayer) is **combinatorial**: the tick-2/3 integer comparison determines the outcome, and spectral methods have no access to that comparison.
2. Reachability contralateral (with or without delayer) is **spectral under scalar- r** with $\rho(A)$ classifier: 98.5% undelayed, 95.8% delayed (both near their respective majority baselines, but with clean ρ -distribution separation in the undelayed case).
3. The delayer-induced winner asymmetry (FCS Fig. 11) is itself combinatorial, arising from the extra-tick-of-latency constant entering the same tick-2/3 comparison.

The headline result from the project does not change. Phase 2 adds a third data point consistent with the Phase 1b/1c diagnosis and confirms that the spectral/combinatorial split is a property of the archetype class, not of any specific topology within it.

Generated from `deq/archetypes/phase2_delayer.py`. Artifacts in `deq/archetypes/results/phase2_*` and `fcs_fig11_*`.